

Month 5: Stateful Persistence, User Interface, and Optimizations

This month focused on database integration, user authentication mapping, web UI callbacks, and global model caching.

Week 1: MongoDB Database and Asynchronous Data Layer

- **Progress Summary:** MongoDB was selected as the database backend. The custom `MongoDataLayer` was developed using the asynchronous motor driver. The database collections (users, threads, steps, elements, feedback) and CRUD methods were implemented to store session histories on non-blocking threads.

Week 2: Silent Authentication and Access Control Mapping

- **Progress Summary:** User session security was established using an auto-login callback (`@cl.header_auth_callback`). Access control logic was implemented in the data layer to map database UUIDs back to the session identifier, preventing user authorization errors during historical chat loads.

Week 3: Conversational User Interface Integration

- **Progress Summary:** The Chainlit web interface was integrated. Sidebar sliders and select menus were added to allow users to adjust parameters (top-k, similarity thresholds, models, temperature). The UI callbacks (`@cl.on_chat_start`, `@cl.on_message`) were implemented, linking messages to the hybrid RAG pipeline.

Week 4: Model Caching, Resumption Lifecycle, and Evaluation

- **Progress Summary:** The model cache was implemented (`_GLOBAL_RETRIEVER`), reducing retriever loading times to under 12 milliseconds and preventing WebSocket timeouts. The `@cl.on_chat_resume` callback was integrated to rehydrate settings and past messages. Final performance evaluations were executed, completing the implementation lifecycle.